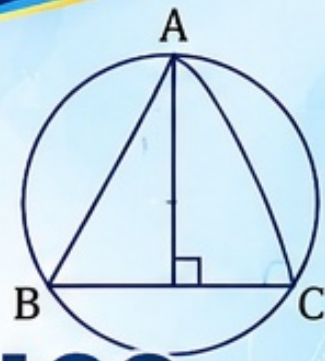
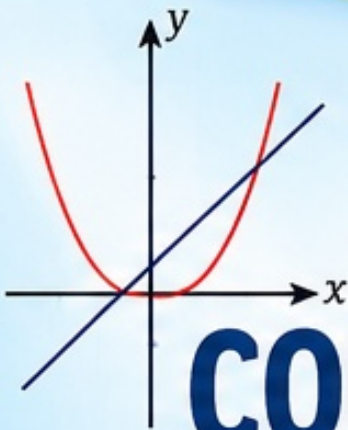




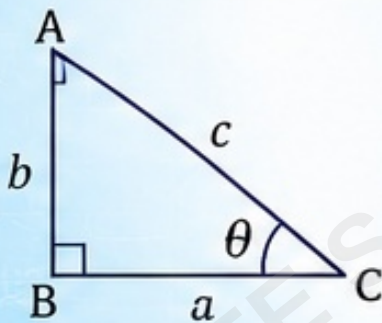
GRADE 11

CORE MATHEMATICS

REVISION BOOK

BOOK 2

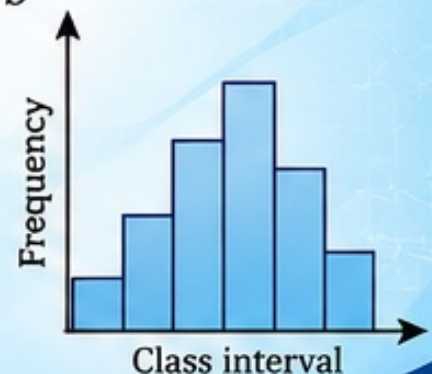
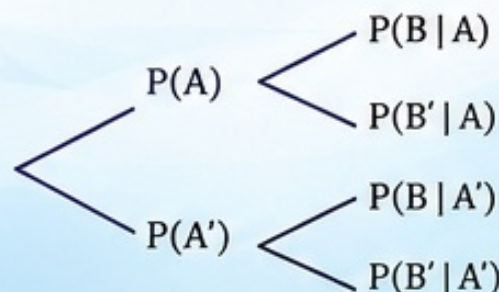
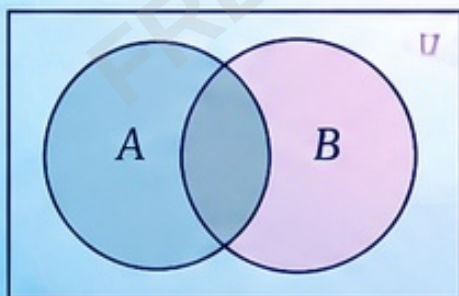
PRACTICE, MASTERY & EXAMINATION



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$



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GRADE 11 CORE MATHEMATICS REVISION

BOOK

BOOK 2 - PRACTICE, MASTERY & EXAMINATION

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Dedication

To the loving memory of my late father, Simeon Nyatuka, whose life, sacrifice and place in our family remain part of the foundation on which I continue to build. Though time moves forward, the lessons, strength and memories you left behind continue to speak. This work is dedicated to your memory with deep love, honour and gratitude.

To Hyrine Kerubo, Peter Munde Mong'are and Teresa Kwamboka Sawe: thank you for the support, guidance, encouragement and sacrifices that have helped shape my journey. Your contribution cannot be measured only in words; it is reflected in every learner I teach, every resource I prepare and every step I take in education.

To Mnaree, Richie, Duke, Caleb and Cuba, and to the friends and companions who have walked with me through demanding seasons: thank you for the conversations, encouragement, laughter, correction and belief. The journey becomes lighter when it is shared with people who genuinely want to see one another grow.

To Mr Peter Kimani and my fellow teachers: thank you for treating teaching as more than the delivery of content. Your professional insight, verification, correction and commitment to strong mathematical thinking remind me that a good revision book must not merely give answers; it must teach a learner how to arrive at them.

To Her, my special one: thank you for the quiet encouragement, patience and presence that give meaning to the hours spent building, teaching and creating. Some support is loud; some is simply steady. Both matter, and yours is deeply appreciated.

Above all, this book is dedicated to every learner who has ever looked at a Mathematics question and wondered whether they could solve it. Book 2 is for the learner who is willing to practise again, correct an error, try a different method and keep going until the reasoning becomes clear. May every worked solution in these pages remind you that Mathematics is not mastered by memorising final answers, but by understanding the steps that lead to them.

May this revision book strengthen your confidence, sharpen your reasoning and prepare you to face unfamiliar problems with patience and courage. Use it actively: attempt the question first, compare your method with the worked solution, identify where your reasoning changed, and then solve another question without looking back.

Education gives us more than marks; it gives us the discipline to think, the courage to improve and the ability to use what we know to build a better future.

- Hezekiah Omweri Nyatuka

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This book is an independently prepared educational revision resource. The notes, worked examples, exercises, Mixed Concepts activities and examination-style practice tests have been developed for learning and revision purposes. References to CBE, national examination style, or similar descriptions indicate the intended competency-based approach and practice format. They do not imply that the questions are official examination papers or that this publication is officially endorsed, approved or published by KNEC or KICD.

How to Use This Book

This is a REVISION AND MASTERY book. Do not use it as a passive notes book. For every set, first attempt the questions without looking at the answer. Then study the worked solution line by line: identify the formula or rule, the substitution, each algebraic or numerical step, and the final answer with its unit. Redo any question you miss. Use the Application Quizzes, Mixed Concepts and six 100-mark practice tests as timed revision. The purpose of Book 2 is repeated practice, correction of errors, examination technique and mastery.

REVISION ROUTINE: Attempt -> Check -> Correct -> Redo -> Time yourself -> Reflect.

For calculation questions, never copy only the boxed answer. Your working should show:
Formula/Rule -> Substitution -> Simplification/Calculation -> Final Answer.

For theory questions, state the mathematical fact or theorem and then apply it to the question.

HINT: A short strategy that helps you choose an efficient method.

REMEMBER: A formula, fact or relationship that should stay in your working memory.

CAUTION: A common error to avoid.

THINK BIG: A prompt that asks you to connect ideas or reason beyond routine substitution.

SPOTLIGHT: A high-value examination or application idea.

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CHAPTER 1 - QUADRATIC EXPRESSIONS AND EQUATIONS

REVISION MODE: This is Book 2 - a practice, mastery and examination revision book. Use the recall points only as a very short formula/theorem reminder. Attempt every question first, then compare your method with the complete worked solution and redo any question you miss.

What is a quadratic equation?

An equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$.

Which methods are used?

Factorisation, completing the square, the quadratic formula and graphical methods.

What does the discriminant show?

For $D = b^2 - 4ac$: $D > 0$ gives two distinct real roots, $D = 0$ one repeated real root, and $D < 0$ no real roots.

REMEMBER: Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

CAUTION: When using a contextual quadratic, reject roots that are impossible in the context.

REVISION PRACTICE - ATTEMPT FIRST, THEN CHECK FULL WORKING

1. Solve $x^2 - 9x + 20 = 0$.

Step-by-step solution:

Step 1 - Form the equation and simplify it systematically.

Step 2 - $(x-4)(x-5)=0$, so $x=4$ or $x=5$.

Final answer: $(x-4)(x-5)=0$, so $x=4$ or $x=5$.

2. Solve $2x^2 + x - 6 = 0$.

Step-by-step solution:

Step 1 - Form the equation and simplify it systematically.

Step 2 - $(2x-3)(x+2)=0$, so $x=3/2$ or $x=-2$.

Final answer: $(2x-3)(x+2)=0$, so $x=3/2$ or $x=-2$.

3. Solve $x^2 + 8x - 9 = 0$ by completing the square.

Step-by-step solution:

Step 1 - Start with $x^2 + 8x - 9 = 0$.

Step 2 - Move the constant: $x^2 + 8x = 9$.

Step 3 - Add $(8/2)^2 = 16$ to both sides: $x^2 + 8x + 16 = 25$.

Step 4 - Factor the perfect square: $(x + 4)^2 = 25$.

Step 5 - Take square roots: $x + 4 = \pm 5$.

Step 6 - Hence $x = 1$ or $x = -9$.

Final answer: $x = 1$ or $x = -9$.

4. Solve $5x^2 - 7x - 3 = 0$.

Step-by-step solution:

Step 1 - Form the equation and simplify it systematically.

Step 2 - $x = [7 \pm \sqrt{109}] / 10 \approx 1.744$ or -0.344 .

Final answer: $x = [7 \pm \sqrt{109}] / 10 \approx 1.744$ or -0.344 .

5. Classify $4x^2 - 4x + 1 = 0$.

Step-by-step solution:

Step 1 - Use the discriminant $D = b^2 - 4ac$.

Step 2 - $D = (-4)^2 - 4(4)(1) = 0$, so repeated root $x = 1/2$.

Final answer: $D = (-4)^2 - 4(4)(1) = 0$, so repeated root $x = 1/2$.

CBE APPLICATION REVISION - ATTEMPT FIRST, THEN CHECK FULL WORKING

6. A rectangle has area 96 m^2 and length 4 m more than width. Find dimensions.

Step-by-step solution:

Step 1 - Let the width be $x \text{ m}$, so the length is $(x + 4) \text{ m}$.

Step 2 - Use area = length $\times$ width: $x(x + 4) = 96$.

Step 3 - Expand: $x^2 + 4x = 96$.

Step 4 - Rearrange: $x^2 + 4x - 96 = 0$.

Step 5 - Factorise: $(x + 12)(x - 8) = 0$.

Step 6 - Thus $x = -12$ or $x = 8$. Reject -12 because a physical length cannot be negative.

Step 7 - Width = 8 m and length = $8 + 4 = 12 \text{ m}$.

Final answer: 8 m by 12 m .

CHAPTER 2 - IRRATIONAL NUMBERS (SURDS)

REVISION MODE: This is Book 2 - a practice, mastery and examination revision book. Use the recall points only as a very short formula/theorem reminder. Attempt every question first, then compare your method with the complete worked solution and redo any question you miss.

What is an irrational number?

A real number that cannot be expressed as a ratio of two integers; its decimal expansion is non-terminating and non-repeating.

What is a surd?

An exact irrational root such as $\sqrt{2}$ or $3\sqrt{5}$.

How is a denominator rationalised?

Multiply numerator and denominator by a suitable surd or conjugate so the denominator becomes rational.

REMEMBER: $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ for non-negative a and b .

CAUTION: Only like surds can be added or subtracted directly.

REVISION PRACTICE - ATTEMPT FIRST, THEN CHECK FULL WORKING

7. Simplify $\sqrt{108}$.

Step-by-step solution:

Step 1 - Express each radical using perfect-square factors; combine like surds or use the conjugate where required.

Step 2 - $\sqrt{(36 \times 3)} = 6\sqrt{3}$.

Final answer: $\sqrt{(36 \times 3)} = 6\sqrt{3}$.

8. Simplify $\sqrt{32} + \sqrt{50}$.

Step-by-step solution:

Step 1 - Express each radical using perfect-square factors; combine like surds or use the conjugate where required.

Step 2 - $4\sqrt{2} + 5\sqrt{2} = 9\sqrt{2}$.

Final answer: $4\sqrt{2} + 5\sqrt{2} = 9\sqrt{2}$.

9. Simplify $5\sqrt{3} - 2\sqrt{3} + \sqrt{12}$.

Step-by-step solution:

Step 1 - Express each radical using perfect-square factors; combine like surds or use the conjugate where required.

Step 2 - $3\sqrt{3} + 2\sqrt{3} = 5\sqrt{3}$.

Final answer: $3\sqrt{3} + 2\sqrt{3} = 5\sqrt{3}$.

Step-by-step solution:

Step 1: This is the perpendicular-to-chord theorem

Step 2: the two right triangles formed have equal radii and common perpendicular, so the half-chords are equal.

Final answer: the two right triangles formed have equal radii and common perpendicular, so the half-chords are equal.

FREE SAMPLE - NOT FOR SALE